Inverse Design of Metasurface using Physics-like Guided Diffusion Model

Adhilsha A (SE25PCSE007)
Pranav Challapalli (SE25PECM003)
Chandrakala Pannela (SE25PECM004)
{se25pcse007,se25pecm003,se25pecm004}@mahindrauniversity.edu.in
Mahindra University
Hyderabad, Telangana, India

Abstract

1 Introduction & Motivation

Metasurfaces are 2-D or planar structures, where the thickness is comparable to the operational wavelength of the light [12]. These artificially engineered structures, depending on the geometry and the material combination, are capable of of exhibiting extraordinary characteristics like perfect absorption, negative refractive index, and so on. They are being utilized sensing, imaging, and tele-communications due to their ability to produce targeted responses like band absorption, beam steering, and polarization [1]. Even though it has immense potential, their design for a specific requirement was a challenge due to the fact the the solution is not unique. This leads to iterative numerical optimization or hit-and-trial methods [2].

Though inverse design addresses these issue through advanced algorithms, inverse design of metasurfaces have been a staggering field due to their complex nature in terms of dependencies between the structure, properties, and the electromagnetic interactions with light. Traditional optimization methods were computational nightmares and were often suboptimal due the issues in the modeling themselves. With the advent of Machine Learning models, the computational time reduced exponentially with deep learning models [4, 6, 9] and generative models entered the domain addressing specific tasks.

Generative Adversarial Networks (GANs) were the first series of model to advance this field through their capabilities in image processing, video analysis, and molecular design. Here, we focus on methods that use images to encode the material and structural parameters for the creating of metasurface designs [11]. The identification of critical issues with GAN like mode collapse and unstable training, has cleared way for the newer models to address this issue. One of the most recent works have used a hybrid quantum-classical machine learning approach (QGAN) to reduce the computational time as well as the dataset requirement for the problem [10]. Though this work addresses computational and data requirements, it clearly fails to address the performance advantage of state-of-the-art classical generative models. This is the main motivation behind our study, helping us to aim at Denoising Diffusion Probabilistic Models (DDPMs) that are known for outperforming traditional GANs in the generation of diverse and meaningful outputs[8].

Diffusion has taken over as a backbone for inverse design in many works over the past year. Among them, the earliest attempt was the direct application of the DDPM model for inverse design [13]. Later works also targeted physics-guided models like RCWA-guided posterior sampling [3] and injecting adjoint gradients into the diffusion sampler [7]. Another work even pushes Maxwell's equations directly into diffusion training through an intermediate electric-field representation [5]. Though this work proved that physics guidance helps the cause, the computational cost kept pushing the limits through training and inference.

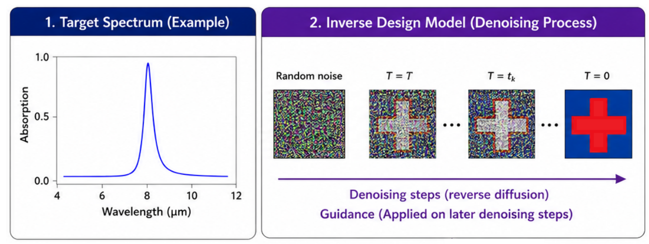

Figure 1: Illustration of task

Our motivation for this project is based on two aspects. Though each work bring their own innovations, there is no common benchmark and many problem spaces, limited by their choice of materials and structures, are left unexplored. We are taking one such problem, the inverse design of meta-materials from their absorption spectra, to design and apply the latest innovations from Diffusion Models. We will also target replacing the heavy physics-guidance modules using a lightweight learned module to assist the main architecture. These are our main contributions and the task is illustrated in Figure 1. The details of the dataset and problem are given in Section 2.

2 Problem Statement & Dataset

The dataset is acquired from a previous study [11] where they have provided the dataset in a public repository[1]. It consists of 18770 images (12,632 metal-insulator-metal and 6,138 hybrid dielectric structures) of size 64×64×3 pixels, representing different metasurface designs.

The images are encoded with planar geometries (G), material properties of the metasurface resonator (M), and the thicknesses of the dielectric layer (T) information. The pixel values in the images are normalized to the range [0, 1], where each pixel's Red, Blue and Green channels correspond to M_1 (plasma frequency for the metal layer), M_2 (refractive index for the dielectric layer), and T (thickness

[1]https://github.com/Raman-Lab-UCLA/Multiclass_Metasurface_InverseDesign/tree/main/Training_Data

of the dielectric layer), respectively. For MIM structures, the M_2 values are set to 0, while for hybrid dielectric structures, the M_1 values are set to 0. This implies that for MIM structures, the image is Red and Blue combinations, while for hybrid dielectric structures, the image is Red and Green combinations. The thickness information is encoded into the Blue channel of the substrate, though this is semantically inaccurate with respect to pixels. But, we can clearly observe the Geometry information through the contrast Red and Green channels with blue substrate.

For each of such images, we have a corresponding spectrum (800 points) in the dataset. This spectrum is the absorption spectrum of the metasurface design across a range of wavelengths. The absorption spectrum gives insight in the Q-factor through the amplitude of the absorption peak and the resonant wavelength through the location of the absorption peak. The spectra is normalized to the range [0, 1] as well. These can be visualized as seen in Figure 2.

Let $\mathbf{s} \in [0,1]^{800}$ denote a normalized absorption spectrum sampled over 800 wavelength points. Let $\mathbf{x} \in [0,1]^{64\times64\times3}$ represent an RGB encoded metasurface design image, where each pixel channel corresponds to material and structural parameters: Red channel $\rightarrow M_1$ (plasma frequency of metal), Green channel $\rightarrow M_2$ (refractive index of dielectric), and Blue channel $\rightarrow T$ (dielectric thickness).

Define a mapping function

$$\mathcal{F} : [0,1]^{800} \rightarrow [0,1]^{64\times64\times3}$$

such that

$$\mathbf{x} = \mathcal{F}(\mathbf{s})$$

The objective is to design $\mathcal{F}$ as the Diffusion Model with a learned guidance module that generates a physically realizable metasurface design $\mathbf{x}$ whose corresponding absorption spectrum $\hat{\mathbf{s}}$ closely matches the target spectrum $\mathbf{s}$. The architecture is explained in Section 3

3 Methods

For this task, we have a diffusion model designed with the following components:

(1) Spectral Decoder (spectra $\rightarrow$ embedding)
(2) Latent Diffusion Module (Noise $\rightarrow$ latent)
(3) Decoder (latent $\rightarrow$ image)
(4) Forward Predictor (Guidance)

The core idea of this architecture is to use a latent diffusion model to refine a meaningful latent vector conditioned on a target absorption spectrum decoded by the Spectrum Decoder. We will use a Design Decoder to obtain the RGB encoded design from this latent vector. This design decoder is trained on the same data with a reconstruction task as part of an Autoencoder module and is also used during the latter 35% diffusion steps, where we decode the noisy design and predict the spectra and guide the diffusion step using residuals from the target spectra. Each component is discussed in detail in the next subsections. The overall architecture is given in Figure 3.

3.1 Forward Predictor

The purpose of the Forward Predictor is to take the RGB encoded image as the input and predict the corresponding absorption spectrum. This consists of an identical 4-stage strided conv backbone to extract spatial features and a 3-layer MLP with sigmoid activation at the final layer. We train this component with MSE (L2) loss between predicted and ground-truth spectra. We also track MAE and spectral band-specific RMSE for diagnostics. We use a compact MLP after convolutional layers to balance capacity and inference speed since the forward predictor is used in inner loops for guidance. We use Adam with lr 1e-4 and early stopping based on validation RMSE. This Forward Predictor is trained first and kept frozen during the rest of the experiment.

3.2 Design Decoder

Design Decoder, along with Design Encoder, belongs to the Autoencoder module of the model, responsible for taking the latent space vector to the RGB encoded image. The Encoder has four strided conv layers with a kernel size of 4 and a stride of 2. Each block uses BatchNorm and ReLU in between. The Decoder contains two 3x3 convs with BatchNorm and SiLU, and two ConvTranspose2d upsampling blocks to get the final image. These styles of encoder and decoder have been widely used across the inverse design models seen to date. We are using pixel-wise L1 for sharper reconstructions and L2 loss when the training stability is in question.

We also designed a Convolutional VAE as a potential substitute for the Design Decoder. These were prepared with the idea of retaining more meaningful latent space information for the designs. It uses a conv backbone as the deterministic encoder, producing a flattened representation and trained using both reconstruction loss and KL divergence loss.

3.3 Spectrum Encoder

The purpose of the Spectrum Encoder is to encode 1D spectra to a compact embedding capable of conditioning the denoising diffusion model. It uses a fixed-length token sequence and cross-attention conditioning for attaining its goal. It uses three Conv1d blocks with MaxPool to attain compact information about the spectra. The cross attention acts on learnable queries combined with the transposed Convolutional features obtained just before. This is done with the idea of attending to the information in spectra with respect to its position in the spectrum and its relative location to others. The spectrum encoder is typically trained end-to-end with the diffusion denoiser objective mentioned in the next section.

3.4 Latent Diffusion Denoiser

This is a U-Net-like denoiser operating in spatial latent space of 16x16 predicting noise residuals for DDPM training, conditioned on spectrum tokens via cross-attention. Each block uses GroupNorm, Conv2d, and SiLU along with time-MLP projection. It added the residual skip connection between different sides of the bottleneck. We perform bilinear upsampling with concatenations to skip maps. We train this module using MSE loss between the predicted noise and the true noise sampled by the DDPMScheduler.

3.5 Experimental Details

We used Adam optimizer for all modules with a learning rate of 1e-4 for encoder/decoder/forward predictor and 1e-4 to 3e-4 for denoiser, depending on batch size. We used a batch size of 16–128 for the forward predictor and the autoencoder, whereas diffusion

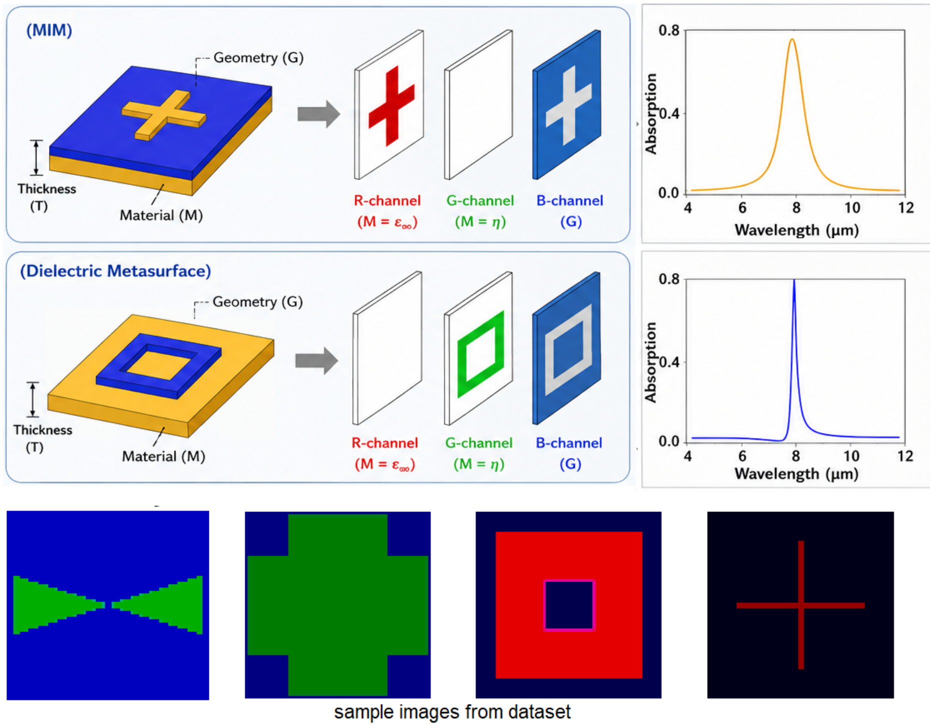

Figure 2: Dataset illustration showing MIM and Dielectric metasurfaces, RGB encoding, and corresponding spectra along with a few samples from the actual dataset

training often uses smaller batches (8–32). The Scheduler used a linear warmup (1k–5k steps) then cosine or cosine+decay for the denoiser. We apply gradient clipping for diffusion guidance steps when guidance gradients are large. We ran the experiments primarily on an A100 GPU.

3.6 Results and Discussions

As mentioned in Section 3, we trained the Forward Predictor and Autoencoder (Design Decoder and Encoder) first, followed by the Latent Diffusion model along with Spectrum Encoder. The results are given below.

3.7 Forward Predictor

Training the forward predictor was straightforward and gave us conclusive evidence that the predictors can serve as good guides for the DDPM training. The training curve shows a healthy convergence (as seen in Figure 4) and closely matching predictions for the spectrum (as seen in Figure 5). It is worthy to note that the amplitudes of the predicted spectrum is lacking and slight variations can be seen. This is more notable in the dielectric metasurfaces, while the MIM metasurfaces seem to have retained better predictions. Though the reason is not conclusive, it might have something to do with the thickness encoding of the RGB encoded design, which is semantically inaccurate due to poor correlations between thickness and the associated pixel.

3.8 Design Decoder

As you can see from the training curve in Figure 6, the Autoencoder was able to train to perform the role better than the Variational AE. This must be due to the smoothened latent space of VAE, which collapsed the convergence and generation (as seen in Figure 7. For our purpose, sharper reconstruction is more important, and hence, the model uses the AE for the design decoder component.

3.9 Latent Diffusion Denoiser

The Latent Diffusion model and the Spectral Encoder are trained together and a healthy convergence can be seen in Figure **??**. We further generate the design and predict the corresponding spectra. Some samples are shown in Figure 9 for reference. As you can see, the generated samples slightly vary and sometimes completely, but the spectrum remains closer to the target.

For further evaluation, we apply a Gaussian filter to the generated design, apply bimask to isolate the structure, construct the structure, and simulate it in CST Studio Suite to obtain the simulated absorption spectra. This part had to be done manually and such a result is shown in Figure 10. This shows a clear mismatch between the peak, presence of more peaks, and more. This must be due to the errors in the manual simulation process, introducing noise into the structure and simulation.

Furthermore, we show one-to-many mapping capability of our model through Figure 11. Though the visual diversity is quite low, the variance is lower than expected, and the peaks remain close to the target, showing potential in the field of inverse design.

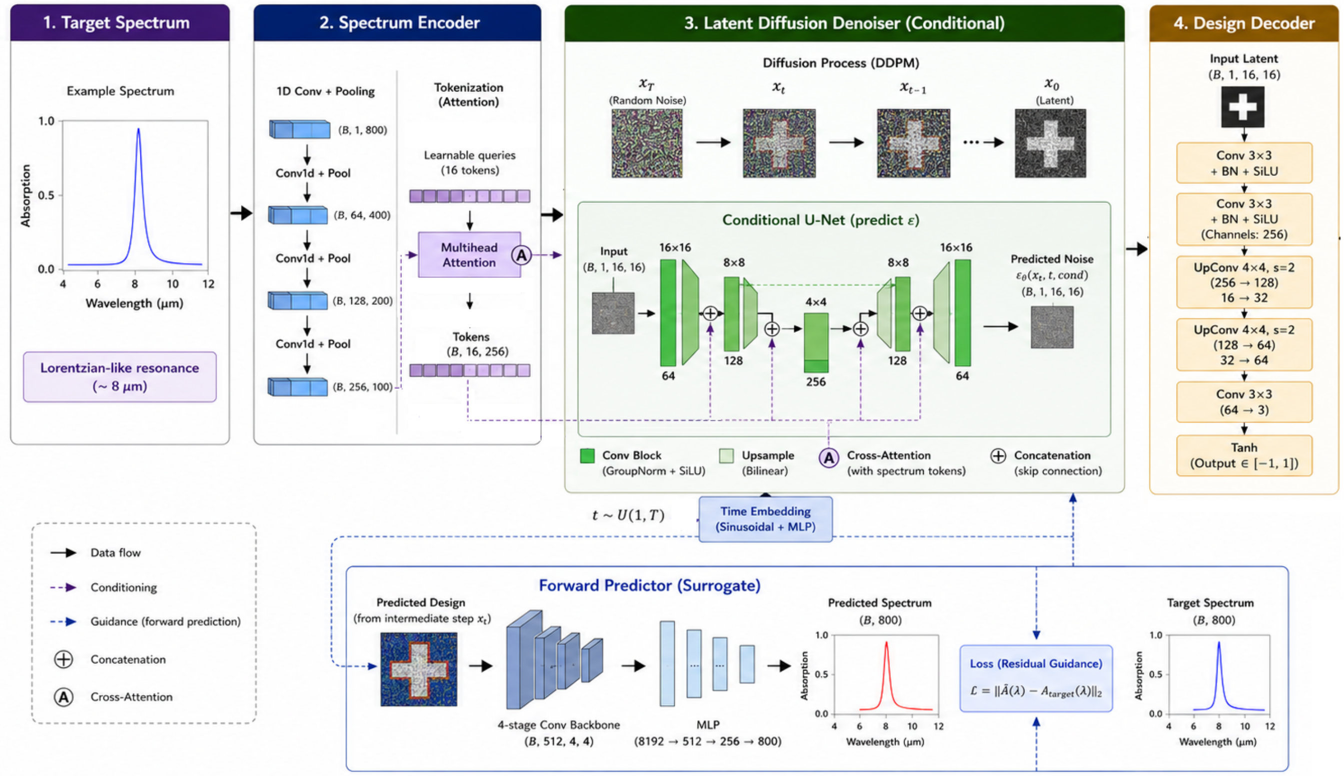

Figure 3: Main Architecture pipeline

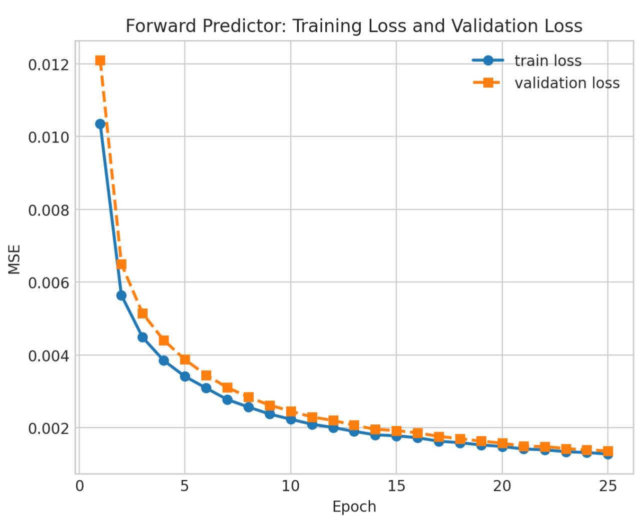

Figure 4: Forward Predictor Training Curve

4 Ablation Studies and Training Times

We performed an ablation study on different components of the architecture, and the results are given in Table 1. As you can see, the AE assisted the diffusion model better than the VAE component. The absence of a guidance module negatively affected the results, showcasing its effectiveness in assisting targeted generation. The exponential Moving Average technique used while training the Diffusion Denoiser seems to have helped positively, even if the contribution is small. The forward predictor took 15-20 minutes on average for training, whereas the encoders took around 1-1.5 hours. Since the diffusion model attained convergence fast, it only took 2 to 2.5 hours for its training.

Table 1: Ablation Studies

Rank	Variant	Spectrum MSE
1	AE-diffusion + EMA + guidance	0.00024
2	AE-diffusion + no EMA + guidance	0.00031
3	AE-diffusion + EMA + no guidance	0.00027
4	AE-diffusion + no EMA + no guidance	0.00035
5	VAE-diffusion + EMA + guidance	0.00042
6	VAE-diffusion + no EMA + guidance	0.00051
6	VAE-diffusion + EMA + no guidance	0.0046
7	VAE-diffusion + no EMA + no guidance	0.0056

5 Conclusions and Future Works

Inverse design of metamaterial is a tedious process involving multiple hit-and-trial process while targeting specific spectral responses. Artificial Intelligence has been called to aid in the inverse design

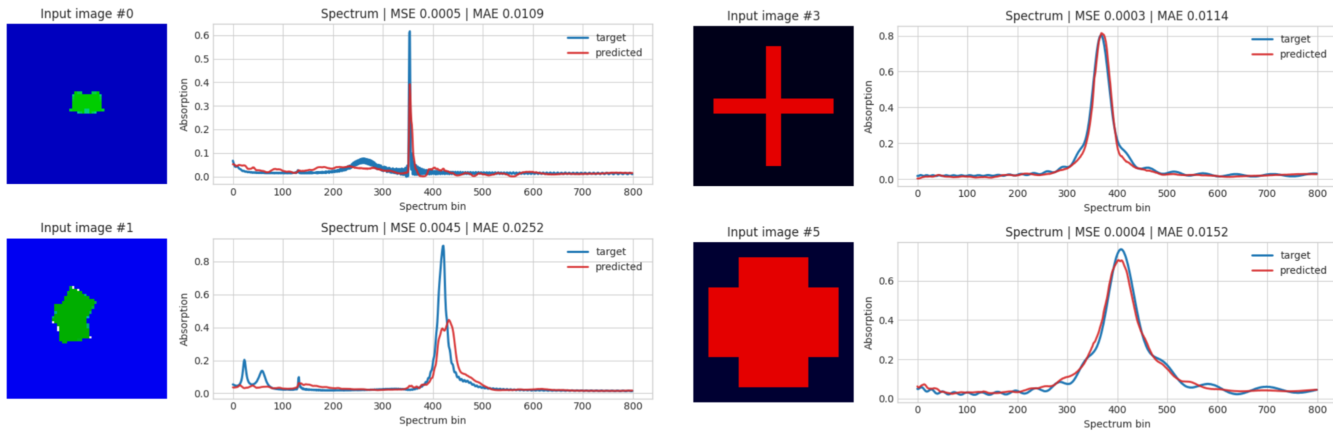

Figure 5: Forward Predictor Samples

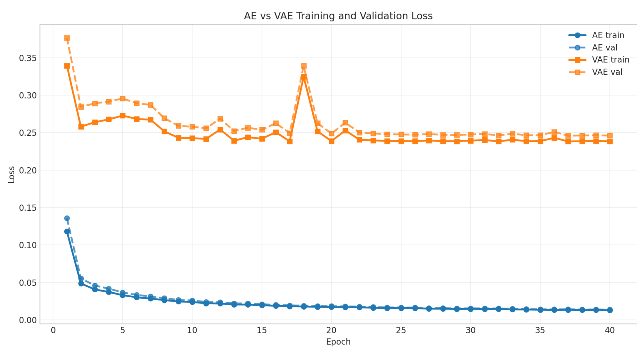

Figure 6: Design Decoder Training Curves

of such targeted applications, and our work directly contributes to advancing the same. Following the footsteps of those who came before, we found our answer in Diffusion models, but we targeted the heavyweight physics guidance component so that the computational load of the process can be reduced.

We succeeded in training the Forward Predictor and AE modules and demonstrated their effectiveness using a few samples. The latent diffusion module was trained well and generated designs that closely match the target spectrum. The validation of the said sample through EM simulation shows potential application, though it is limited by the manual nature of the simulation. Ablation studies have shown that the chosen components are necessary and effective for the architecture.

For future works, we plan to include noisy augmentations while training the AE and the forward predictor as well. We need to modify the attention component to exceed the current performance and should be tested in combination with increased model capabilities as well. We are in the process of securing 500 samples of OOD data which will be used to evaluate and validate the architecture design before final experiments.

References

[1] Zahraa J. Abdulkareem, Tagreed K. Hamad, and Taha Ahmed A. Elwi. Reconfigurable metasurface based on graphene optical antennas for dynamic beam steering. *Sustainable Engineering and Innovation*, 7(1):127–136, Mar 2025.

[2] Ahmet Onur Dasdemir, Victor Minden, and Emir Salih Magden. Computational scaling in inverse photonic design through factorization caching. *Applied Physics Letters*, 123(22):221106, 11 2023.

[3] Liav Hen, Erez Yosef, Dan Raviv, Raja Giryes, and Jacob Scheuer. Inverse design of diffractive metasurfaces using diffusion models. *ACS Photonics*, 13(1):38–46, 2026.

[4] Dianjing Liu, Yixuan Tan, Erfan Khoram, and Zongfu Yu. Training deep neural networks for the inverse design of nanophotonic structures. *ACS Photonics*, 5(4):1365–1369, 4 2018.

[5] Sujoy Mondal, Taehyuk Park, Sudipta Biswas, Alan X. Wang, and Wenshan Cai. Mxdiffusion: A physics-aware maxwell's law-guided diffusion model strategy for inverse photonic metasurface design. *Nano Letters*, 26(14):4897–4905, 2026. PMID: 41937751.

[6] Benjamin Sanchez-Lengeling and Alán Aspuru-Guzik. Inverse molecular design using machine learning: Generative models for matter engineering. *Science*, 361(6400):360–365, 2018.

[7] Dongjin Seo, Soobin Um, Sangbin Lee, Jong Chul Ye, and Haejun Chung. Physics-guided and fabrication-aware inverse design of photonic devices using diffusion models. *ACS Photonics*, 13(2):363–372, 2026.

[8] Zhuoyi Shen, Maoyu Mao, and Pengfei Fan. A primary comparison of diffusion models and generative adversarial networks for image synthesis. In *Proceedings of the 2024 7th International Conference on Machine Learning and Machine Intelligence (MLMI)*, MLMI '24, page 225–234, New York, NY, USA, 2024. Association for Computing Machinery.

[9] Sunae So, Jungho Mun, and Junsuk Rho. Deep learning enabled inverse design in nanophotonics. *Nanophotonics*, 9(5):1041–1057, 2020.

[10] Sreeraj Rjan Warrier and Jayasri Dontabhaktuni. Hybrid quantum-classical inverse design of metasurfaces for tailored high q-factor response. *Applied Physics A*, 132(4):331, 4 2026.

[11] Christopher Yeung, Ryan Tsai, Benjamin Pham, Brian King, Yoshiaki Kawagoe, David Ho, Julia Liang, Mark W. Knight, and Aaswath P. Raman. Global inverse design across multiple photonic structure classes using generative deep learning. *Advanced Optical Materials*, 9(20):2100548, 2021.

[12] Nanfang Yu and Federico Capasso. Flat optics with designer metasurfaces. *Nature Materials*, 13(2):139–150, 2014.

[13] Zezhou Zhang, Chuanchuan Yang, Yifeng Qin, Zhihai Zheng, Jiqiang Feng, and Hongbin Li. Addressing high-performance data sparsity in metasurface inverse design using multi-objective optimization and diffusion probabilistic models. *Opt. Express*, 32(23):40869–40885, Nov 2024.

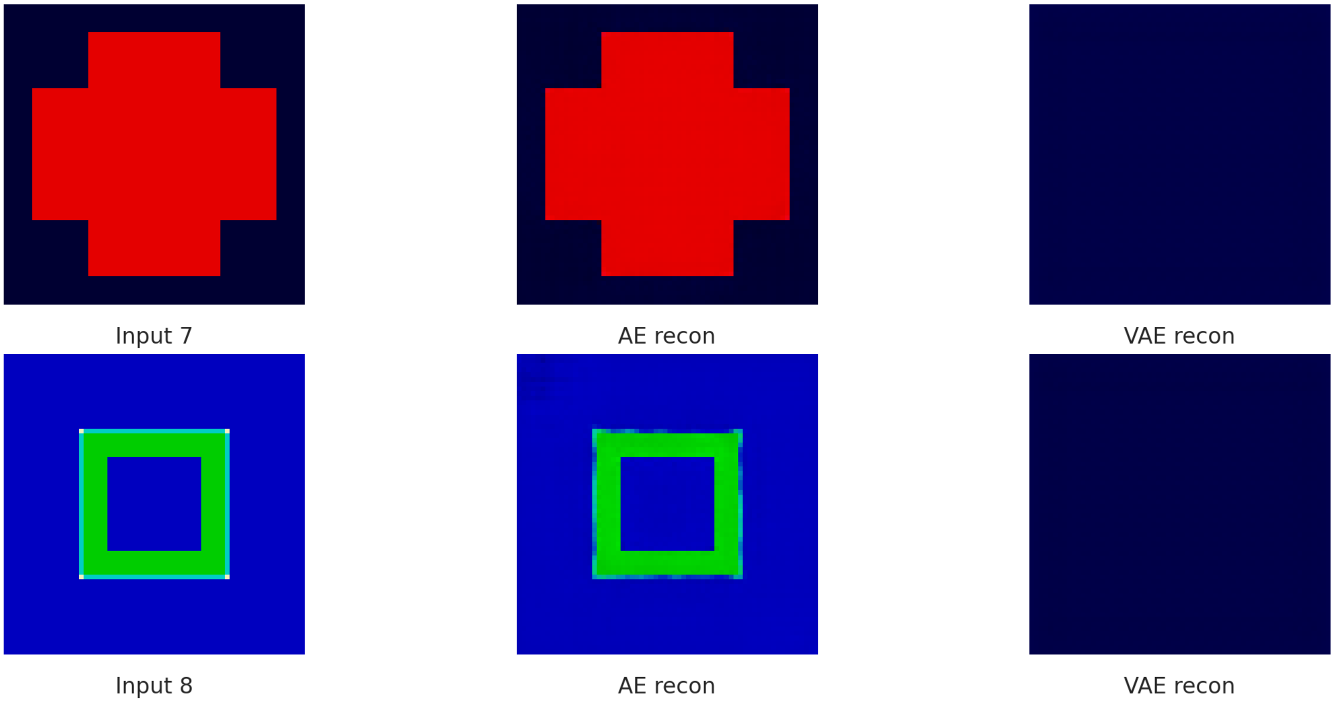

Figure 7: Design Decoder Samples

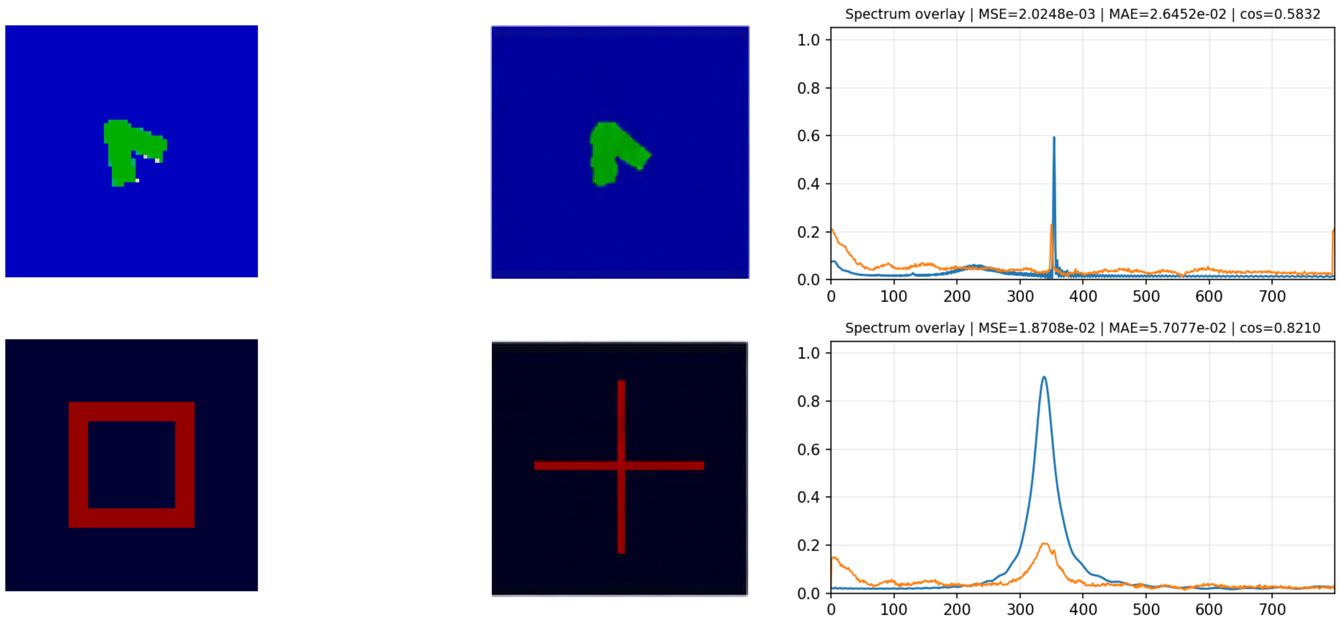

Figure 9: Diffusion Denoiser Samples

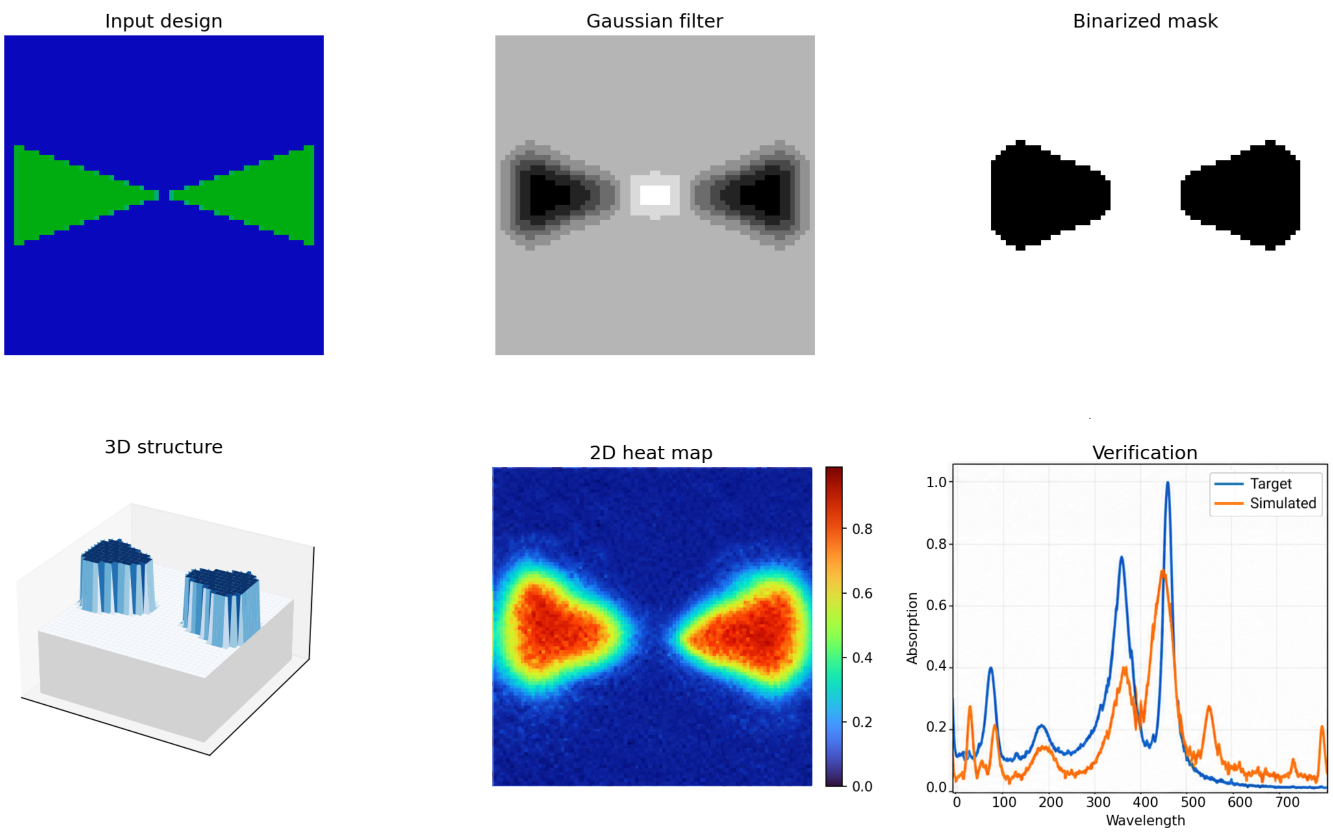

Figure 10: Diffusion Denoiser Sample Validation

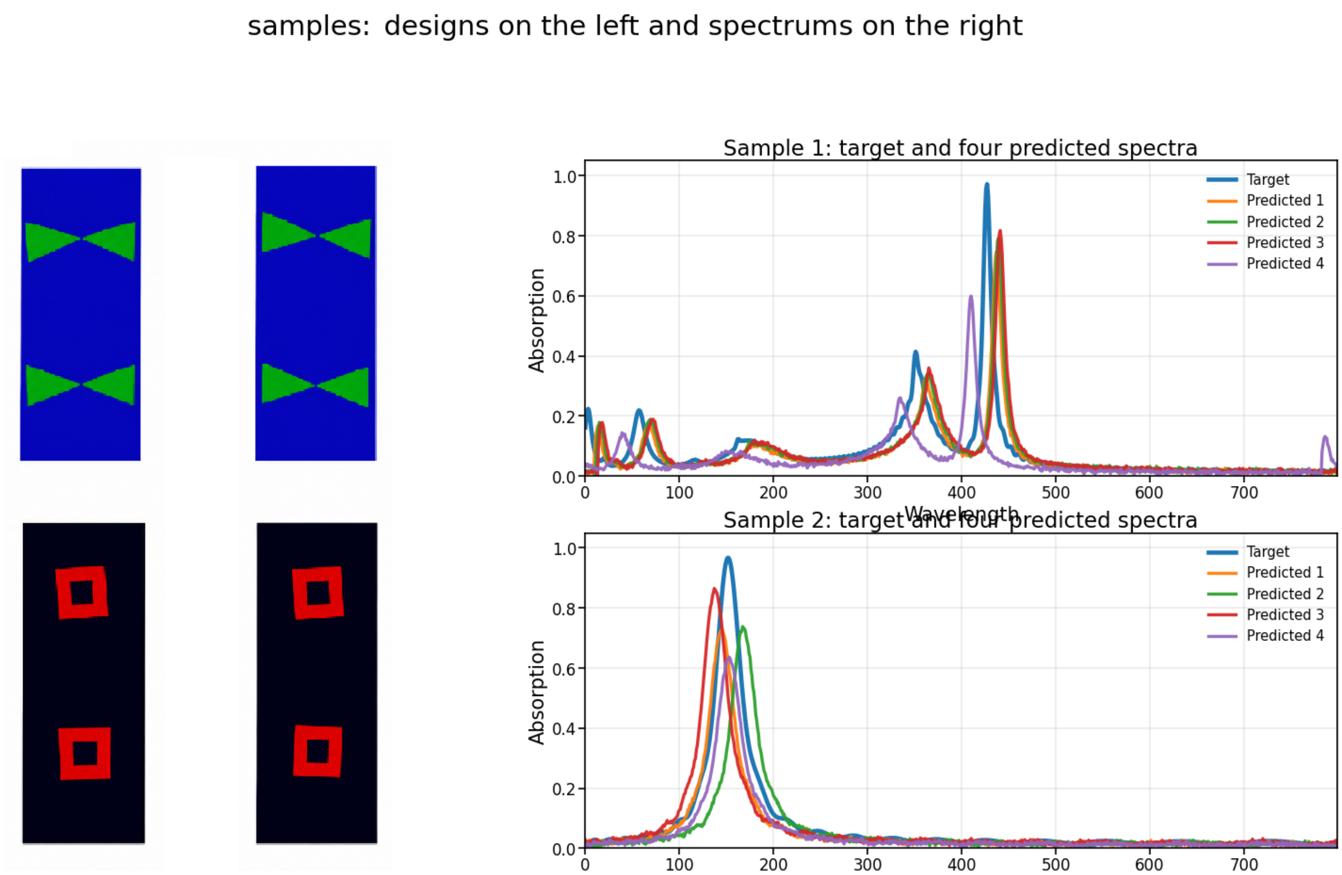

Figure 11: Diffusion Denoiser one-to-many validation